

IILM UNIVERSITY, GREATER NOIDA

School of Computer Science and Engineering

Summer Internship Report Presentation (AY 2026-27)

INTERNSHIP TOPIC:

Cloud Infrastructure Automation & Containerized Deployment Platform

AWS • Terraform • Docker • Kubernetes • Jenkins CI/CD • Prometheus & Grafana

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1. Executive Summary & Internship Context

1.1 Project Scope & Purpose

Architected and validated an automated, cloud-native application delivery platform on Amazon Web Services (AWS) using Infrastructure as Code (IaC) and container orchestration for enterprise reliability.

1.2 Key Technical Accomplishment

Formulated modular HashiCorp Terraform templates to provision multi-AZ AWS Virtual Private Clouds (VPC), EC2 nodes, Application Load Balancers (ALB), and Amazon RDS instances with 100% automation.

1.3 CI/CD Pipeline Automation

Constructed automated continuous integration and delivery pipelines using Jenkins and GitHub Actions, incorporating automated unit testing, security vulnerability scanning, and zero-downtime rolling deployments.

1.4 Operational Observability

Integrated Prometheus time-series metric collection with Grafana dashboards and Alertmanager triggers, reducing Mean Time to Resolution (MTTR) by over 50%.

2. Organization Profile & Internship Role Scope

Organization Overview: SystemaOps

- Specialized Cloud Engineering & DevOps consultancy (systemaops.com).
- Assists organizations in transitioning from monolithic legacy systems to cloud-native microservices.
- Delivers end-to-end solutions in AWS Cloud, Kubernetes orchestration, Site Reliability Engineering (SRE), and DevSecOps.
- Prioritizes infrastructure security compliance, high availability, and automated disaster recovery.

Internship Role & Technical Scope

- Served as a Cloud & DevOps Engineering Intern for 3 months (Summer 2026).
- Provisioned modular Terraform IaC templates for AWS VPC subnets, Application Load Balancers, and EC2 nodes.
- Containerized multi-tier application workloads using Docker and Docker Compose.
- Constructed automated Jenkins and GitHub Actions CI/CD pipelines.
- Implemented Prometheus metrics collection and Grafana operational dashboards.

3. Problem Statement & Project Objectives

3.1 Identified Operational Challenges

- **Environmental Inconsistency:** Discrepancies between local development, testing, and production environments leading to software defects.
- **Manual Cloud Configuration:** Manual server setup causing configuration drift, human error, and prolonged deployment cycles.
- **Deployment Downtime:** Absence of automated continuous deployment leading to service outages during releases.
- **Inadequate Operational Telemetry:** Lack of centralized metric tracking delaying incident root-cause diagnosis.

3.2 Technical Project Milestones

- **Milestone 1:** Formulate modular Infrastructure as Code (Terraform) to provision Multi-AZ AWS VPC, ALB, EC2, and RDS.
- **Milestone 2:** Containerize full-stack application microservices using multi-stage Dockerfiles and Docker Compose.
- **Milestone 3:** Construct automated CI/CD pipelines in Jenkins and GitHub Actions for continuous testing and deployment.
- **Milestone 4:** Configure Kubernetes pod auto-scaling, self-healing, and Nginx reverse proxy ingress routing.
- **Milestone 5:** Deploy Prometheus & Grafana monitoring stack for real-time telemetry and alerting.

4. Multi-AZ AWS 3-Tier System Architecture

Tier 1: Public Presentation Tier (AWS ALB & Nginx)

AWS Application Load Balancer (ALB) distributed across Public Subnets across multiple Availability Zones (Multi-AZ). Manages external HTTPS traffic on Ports 80/443, SSL/TLS termination, and ingress routing to private application nodes.

Tier 2: Private Application Tier (EC2 & Kubernetes Cluster)

EC2 Auto-Scaling Group and Kubernetes Worker Nodes isolated inside Private Subnets. Hosts containerized microservices on Ports 3000/5000. Outbound internet access for software security updates is managed via AWS NAT Gateways.

Tier 3: Private Data Tier (Amazon RDS Database)

Amazon Relational Database Service (RDS - Multi-AZ MongoDB/PostgreSQL) isolated in Private Database Subnets. Features automated multi-AZ storage replication, automated snapshots, and strict Security Group ingress rules.

5. Enterprise DevOps Stack & Infrastructure as Code

Category	Technology / Tool	Technical Role & Implementation
Cloud Infrastructure	Amazon Web Services (AWS)	Host for VPC subnets, EC2 nodes, ALB, S3, RDS, and CloudWatch.
Infrastructure as Code	HashiCorp Terraform	Declarative provisioning of network topology with S3 remote state lock.
Containerization	Docker & Docker Compose	Multi-stage lightweight image builds and multi-container local testing.
Container Orchestration	Kubernetes & Helm	Pod scaling, self-healing, rolling updates, and Helm release management.
CI/CD Automation	Jenkins & GitHub Actions	Automated code checkout, unit testing, image tagging, ECR push, and rollout.
Observability & Metrics	Prometheus & Grafana	Time-series metric collection, target scraping, and dashboard visualization.

6. Containerization & Kubernetes Orchestration

6.1 Docker Container Optimization

- Multi-stage Dockerfiles separating build dependencies from production execution runtime.
- Container image size reduced from 850 MB to 180 MB (> 75% footprint optimization).
- Standardized multi-container orchestration via Docker Compose for local development replication.
- Environment configuration variables isolated securely via Docker Secrets.

6.2 Kubernetes Deployment Strategy

- Kubernetes Deployment manifests configured with rolling update strategy (maxSurge=1, maxUnavailable=0).
- Horizontal Pod Autoscaler (HPA) configured for dynamic pod scaling based on CPU threshold (> 70%).
- Liveness and Readiness health probes configured for self-healing pod recovery.
- Nginx Ingress Controller managing SSL/TLS termination and path-based request routing.

7. Automated CI/CD Pipeline Workflow

Stage 1: Source Control Commit

Developers commit code to GitHub feature branches, triggering automated webhook events in Jenkins & GitHub Actions.

Stage 2: Continuous Testing & Scanning

Automated execution of Jest unit tests, code linting, and Trivy static container vulnerability security scans.

Stage 3: Container Image Build

Multi-stage Docker build executed automatically, creating reproducible production container images.

Stage 4: Registry Push to Amazon ECR

Tagging container image with unique git commit hash and pushing to private Amazon ECR repository.

Stage 5: Zero-Downtime Deployment

Executing kubectl rolling update across Kubernetes cluster nodes ensuring 99.99% application availability.

8. Operational Observability & Security Controls

8.1 Prometheus & Grafana Monitoring

- Prometheus time-series scrapers configured to collect metric targets across cluster nodes.
- Node Exporter and kube-state-metrics integrated for operational system tracking.
- Grafana dashboards deployed for real-time visualization of CPU, Memory, and Request Latency.
- Alertmanager configured with automated triggers for threshold breaches (HTTP 5xx rate spikes).

8.2 Cloud Security & Compliance

- AWS IAM Roles enforcing Least Privilege policies across all EC2 and EKS nodes.
- Private Subnet isolation for Amazon RDS database (zero direct public IP exposure).
- Nginx SSL/TLS Termination utilizing modern TLS 1.3 encryption protocols.
- JWT Token Authentication & Role-Based Access Control (RBAC) in application backend.

9. Quantified Results, Conclusion & Deliverables

9.1 100% Infrastructure Automation

Cloud infrastructure provisioning timeline reduced from hours to under 8 minutes via HashiCorp Terraform modules.

9.2 99.99% Application Uptime

Zero-downtime continuous deployment achieved through Kubernetes rolling update strategy.

9.3 50%+ MTTR Reduction

Prometheus & Grafana alerting accelerated root-cause detection and diagnostic response timeline.

9.4 Complete Deliverables Package

All code artifacts, Terraform modules, report PDF, presentation PDF, and certificate PDF pushed to public GitHub repository.